



Machine Learning Project

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1 Abstract

The aim of this study is to evaluate the effectiveness of different algorithms in building a predictive classifier that can support the management of a mobile phone operator in forecasting customer churn using an operational dataset. The analysis compares a range of models from both Statistical Learning and Machine Learning. Our modelling strategy is explicitly based on testing multiple algorithms, in line with the “no free lunch” principle of machine learning, according to which no single model is universally superior and performance is inherently dependent on the specific characteristics of the dataset. A preliminary exploratory analysis of the dataset was conducted and statistical tests were used to assess the assumptions and hypotheses underlying the models. To improve interpretability, feature importance was investigated and SHAP values were employed to quantify the contribution of each predictor to individual churn predictions. Finally, a fairness analysis of the models was carried out with respect to gender, computing group fairness metrics based on the criteria of Independence, Separation, and Sufficiency. The results provide insights into the trade-offs between accuracy, interpretability, and fairness in churn prediction models for telecom applications.

2 Exploratory Data Analysis

The analysis is based on the dataset `celldata.csv`, which contains $n = 8000$ observations. For each customer we observe

- the target variable **Churn**, and
- $p = 10$ covariates: **CreditScore**, **Geography**, **Gender**, **Age**, **Tenure**, **Balance**, **NumOfProducts**, **HasCrCard**, **IsActiveMember**, **Salary**.

The target variable **Churn** is binary, taking value 0 for non-churning customers and 1 for churning customers. The empirical distribution of **Churn** is visualised in Figure 1 by means of a histogram, which shows the proportion of customers who leave the company versus those who stay.

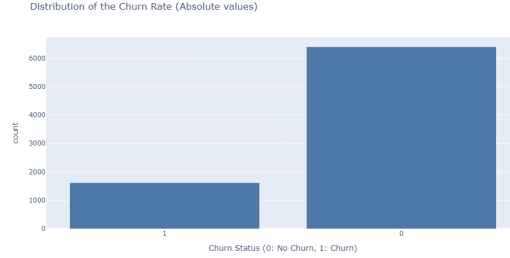


Figure 1: Empirical distribution of the target variable **Churn**.

We assessed the normality assumption required by some of the models by applying a set of statistical tests (Shapiro–Wilk, Kolmogorov–Smirnov, and Q–Q plots) to selected features. Furthermore, since Linear Discriminant Analysis (LDA) relies on the assumption of homoscedasticity, we evaluated the equality of covariance matrices across classes using Box’s M test. We also report below the correlation matrix of the features to assess the independence. In order to better capture potential non-linear dependencies, the correlation matrix was computed using both Kendall’s τ and Spearman’s rank correlation coefficients.

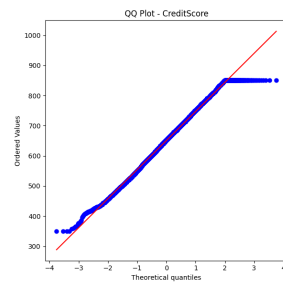


Figure 2: Q-Q Plot - CreditScore

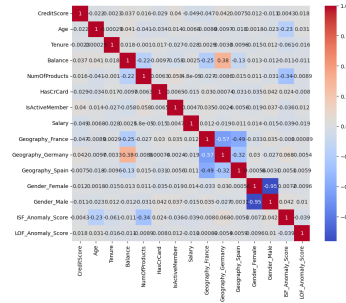


Figure 3: Correlation matrix of the predictors.

2.1 Anomaly Detection

We randomly split the original dataset into training and test sets before any further processing. We then performed anomaly detection using two unsupervised models: Isolation Forest (IF) and Local Outlier Factor (LOF). The anomaly detection models were fitted exclusively on the training set, and the resulting scoring functions were subsequently applied to the test sets, so as to avoid any form of data leakage. For each observation, we computed the anomaly scores produced by IF and LOF and appended them to the original dataset as two additional features; these enriched representations were then used as inputs in the subsequent predictive modelling phase. To make the features comparable in scale and to stabilise the optimisation of the models, we applied z -score standardisation: the mean and standard deviation were computed on the training set only and then used to transform the validation and test sets, again preventing data leakage. Moreover, categorical variables such as **Geography** and **Gender** were encoded using One-Hot encoding.

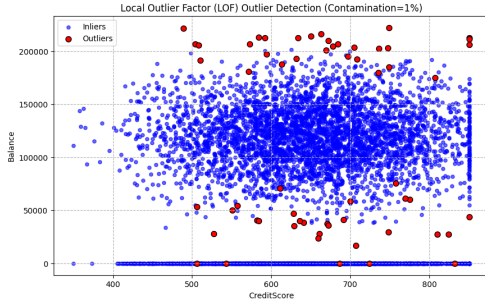


Figure 4: Example of anomaly detection for a single feature.

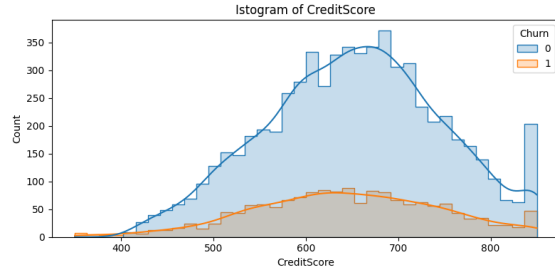


Figure 5: Distribution of CreditScore with respect to Churn

2.2 Feature Engineering & Data Manipulation

The exploratory data analysis revealed a strong imbalance in the target variable **Churn**. To mitigate this issue, we performed a rebalancing of the training set by generating synthetic minority-class observations using the Borderline-SMOTE technique. This method focuses on minority instances that are difficult to classify (e.g., misclassified by a k -nearest neighbours classifier) and oversamples primarily those borderline examples, thereby increasing the resolution of the decision boundary where it is most needed. We use this procedure to balance our training set, while keeping the test set unchanged so as to reflect the real-world class imbalance. Without an appropriate rebalancing strategy, global performance measures such as accuracy can be highly misleading, yielding apparently high values that merely reflect the predominance of the majority class; in such cases, the model fails to effectively learn and detect the minority class corresponding to churners.

3 Statistical Learning Approach

3.1 Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is an effective classification technique, especially suited to problems where the classes can be reasonably modelled as Gaussian distributions sharing a common covariance matrix. Based on the investigations carried out in the EDA phase, we decided to apply a Yeo–Johnson transformation to the variables **Balance** and **CreditScore**, a procedure similar to the Box–Cox transform but applicable also when the data are not strictly positive. Nonetheless, it is worth noting that neither the normality nor the homoscedasticity assumptions are fully satisfied by our data.

Model	Accuracy	Precision	Recall
LDA	0.745	0.4133	0.6366

Table 1: Performance metrics for the LDA classifier.

3.2 Quadratic Discriminant Analysis

Quadratic Discriminant Analysis (QDA) generalises LDA by allowing each class to have its own covariance matrix, thus offering greater flexibility and potentially improved performance when the homoscedasticity assumption is not met. QDA is particularly suitable for problems characterised by non-linear decision boundaries. As in the LDA setting discussed above, we applied a Yeo–Johnson transformation to the variables **Balance** and **CreditScore** prior to fitting the model.

Model	Accuracy	Precision	Recall
QDA	0.7525	0.4332	0.7453

Table 2: Performance metrics for the QDA classifier.

3.3 Logistic Regression

Logistic regression is a classical and widely adopted method for tackling binary classification tasks. It models the probability of a binary outcome as a function of one or more predictor variables, assuming a linear relationship between the covariates and the log-odds of the response, and is appreciated for its simplicity and interpretability. In our implementation, we also included an ℓ_2 (Ridge) penalty on the coefficients, which helps to control overfitting and mitigate multicollinearity by shrinking the parameter estimates towards zero while retaining all predictors in the model. Moreover, we selected the optimal decision threshold using Youden’s index.

Model	Accuracy	Precision	Recall
LR	0.7318	0.4011	0.6739

Table 3: Performance metrics for the LR.

4 Machine Learning Approach

For each of the models considered, we performed hyperparameter optimization via grid search combined with Stratified K-Fold cross-validation, in order to obtain performance estimates that are robust and not overly sensitive to the initial random split of the dataset. Given the relatively small number of features, we did not apply feature selection or dimensionality reduction techniques such as PCA. Once the optimal hyperparameter configuration was identified, we determined the corresponding optimal decision threshold by minimizing Euclidean distance to the optimal point (0,1) in ROC space.

4.1 Decision Tree Classifier

Given the irregular structure of the dataset, we complement the previous parametric models with a more flexible, data-driven approach by introducing a Decision Tree Classifier. We tune its main complexity parameters using the aforementioned validation strategy and select the configuration achieving the highest AUC on the validation set. The final selected tree is then evaluated on the held-out test set, which is kept strictly separate and used exclusively for the assessment of generalization performance.

Model	AUC Score	Accuracy	Precision	Recall
DT	0.7960	0.7831	0.4740	0.7080

Table 4: Performance metrics for the DT classifier.

4.2 Bagging

Bagging builds an ensemble from a single base learner by training multiple models on bootstrap samples and averaging their predictions. In our analysis, we apply Bagging to each weak learner to increase robustness to sampling variability and improve predictive performance, especially for high-variance models like decision trees, at the cost of higher computation and reduced interpretability compared to a single model.

Model	AUC Score	Accuracy	Precision	Recall
Bag LDA	0.76	0.7331	0.4029	0.6770
Bag QDA	0.82	0.7456	0.4271	0.7732
Bag LR	0.76	0.7356	0.4056	0.6739
Bag DT	0.83	0.7525	0.4368	0.7950

Table 5: Performance metrics for the Bagging models.

4.3 Random Forest

Trying to improve classification accuracy and robustness, we train a Random Forest. This model combines multiple decision trees and, in doing so, enhances generalization and reduces the risk of overfitting. However, a key drawback of Random Forest is the loss of interpretability compared to a single decision tree. To address this issue, we compute SHAP

values to quantify the contribution of each feature to the model’s predictions. This step is particularly important in a business context, where it is crucial to explain the main drivers behind the decisions produced by the model.

Model	AUC Score	Accuracy	Precision	Recall
RF	0.8529	0.7918	0.4893	0.7826

Table 6: Performance metrics for the RF classifier.

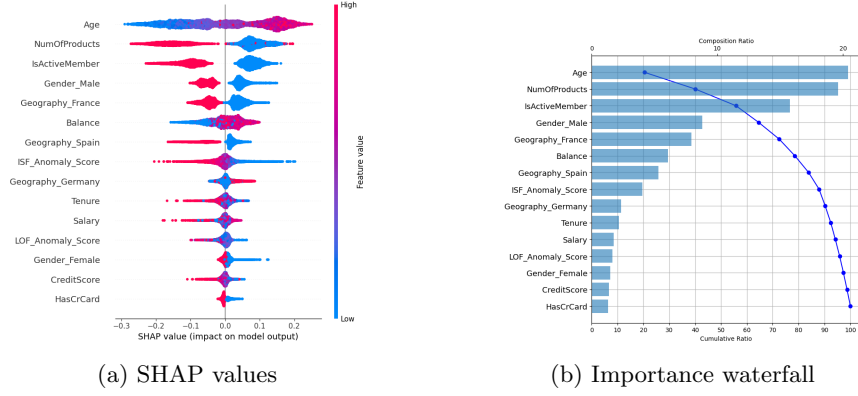


Figure 6: Model interpretability plots: SHAP values and importance waterfall.

4.4 Extra Trees

The Extra Trees method is another ensemble learning algorithm that belongs to the bagging family, similar to Random Forest. It fits a large number of decision trees by randomly choosing variables and thresholds to a dataset and averages their predictions (or uses a majority vote) to improve predictive accuracy and reduce overfitting.

Model	AUC Score	Accuracy	Precision	Recall
ET	0.8281	0.7412	0.4240	0.7981

Table 7: Performance metrics for the ET classifier.

4.5 AdaBoost

AdaBoost combines multiple weak learners into a strong classifier by iteratively reweighting the training instances, placing greater emphasis on those that were misclassified in previous rounds. In this way, the ensemble progressively focuses on harder cases, with the aim of improving overall accuracy and robustness at each iteration. In our experiments, AdaBoost is currently the model that generalizes best: it achieves an AUC of 0.8844 under cross-validation and 0.8611 on the held-out test set, with a much smaller gap between validation and test performance than observed for the other models.

Model	AUC Score	Accuracy	Precision	Recall
AB	0.8611	0.8025	0.5061	0.7639

Table 8: Performance metrics for the AB classifier.

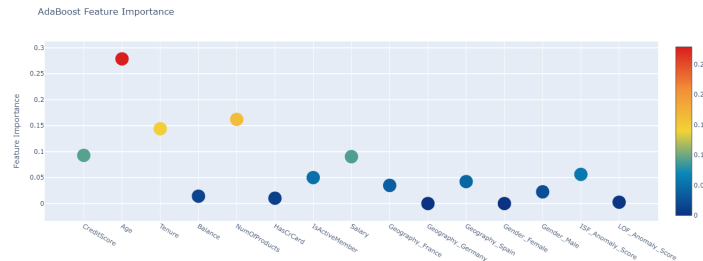


Figure 7: Feature importance for the AdaBoost classifier.

To better understand the reasons for this favourable generalization behaviour, we inspected the feature importance derived from the AdaBoost model. The resulting plot shows that AdaBoost assigns comparatively higher importance to **Tenure** and **CreditScore** than other models such as the Random Forest, as can also be observed from the SHAP analysis of the Random Forest.

4.6 Gradient Boosting

Gradient Boosting constructs models sequentially, where every new model is trained to minimize the residual errors of its predecessors. By aggregating the contributions of multiple weak learners, the algorithm significantly enhances predictive performance and concurrently reduces model bias. This technique demonstrates high efficacy across both regression and classification problems, especially when handling complex datasets.

Model	AUC Score	Accuracy	Precision	Recall
GB	0.8659	0.7812	0.4744	0.8074

Table 9: Performance metrics for the GB classifier.

4.7 XGBoost, CatBoost, LightGBM

XGBoost, LightGBM, and CatBoost represent the modern evolution of the Gradient Boosting framework, focusing on maximizing efficiency and scalability.

Model	AUC Score	Accuracy	Precision	Recall
LGBM	0.8662	0.8293	0.5573	0.7391
XGB	0.8686	0.7968	0.4969	0.7701
CATB	0.8718	0.7975	0.4980	0.7888

Table 10: Performance metrics.

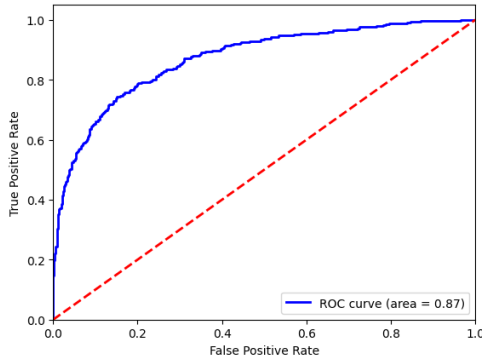


Figure 8: ROC Curve CAT

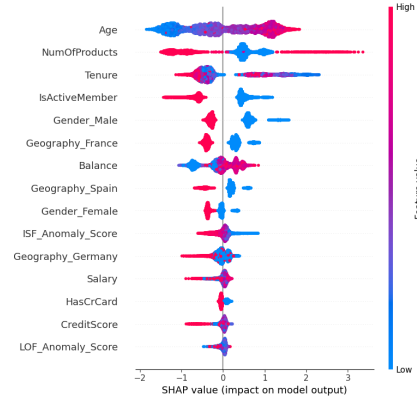


Figure 9: SHAP values CAT

4.8 Summary

Most of the ML models we tried have performances that are comparable with little differences. Among the candidate models, we recommend AdaBoost as the preferred solution not because it reaches the absolute best performance, but because of its strong generalization ability: unlike the other models, which showed AUC gaps of about 6–7 percentage points between cross-validation and test, AdaBoost delivered highly consistent results across validation and test sets. That said, other models should still be considered depending on which metric is most relevant for the business. For example, a higher recall may be desirable if minimizing false negatives is critical, and it is acceptable to contact some customers who would in fact have stayed (**Churn 0**) because the model predicted **Churn 1**. Conversely, in countries with strict regulations on marketing practices, such an aggressive strategy might be inappropriate, and another metric could be prioritized over recall. Additional techniques, such as stacking ensembles and a multilayer perceptron, were also explored but did not yield substantial improvements over the best tree-based ensembles. Furthermore, we report a plot that summarizes the behaviour of the predictors across the four implemented ensemble models by displaying the average of their feature

importances. Finally, we analysed the correlation between the predictions of the different models, finding very strong relationships, with pairwise correlations close to 0.80.

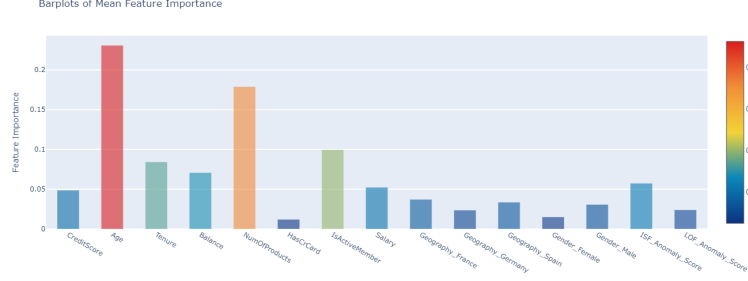


Figure 10: Mean feature importance

5 Fairness Assessment in Classification Models

A pervasive challenge in machine learning is the potential for models to exhibit *bias* towards specific features, often leading to discriminatory outcomes across distinct groups. When sensitive attributes, such as **Geography** or **Gender**, are implicated, this bias translates into a failure to provide equitable classification outcomes. To rigorously evaluate and quantify this issue, several fairness metrics are employed. In the context of this project, we assessed the model’s fairness with respect to the **Gender** feature by calculating three criteria: Independence (or Demographic Parity), *Separation* (evaluated via the ratio of True Positive Rates or False Positive Rates), and *Sufficiency* (assessed via the ratio of Positive Predictive Values or Negative Predictive Values). These ratios are calculated specifically as MetricFemale/MetricMale. For a model to be considered *fair* under these definitions, the calculated metrics should ideally approximate a value of 1.

Model	Independence	Separation TPR	Separation FPR	Sufficiency PPV	Sufficiency NPV
AB	0.5863	0.8865	0.5292	0.9952	1.020

Table 11: Fairness metrics

Initial evaluation indicated that the model did not satisfy the predefined equity thresholds. Consequently, we adopted an intervention strategy, often termed “fairness through unawareness” by training a new iteration of the model after excluding the **Gender** feature from the training set. The objective was to ascertain if this removal would mitigate direct bias while preserving high predictive performance. The results confirmed this approach’s efficacy: performance metrics remained robust (e.g., the AUC Score achieved **0.8579**) and an improvement in the fairness metrics was observed.

Model	Independence	Separation TPR	Separation FPR	Sufficiency PPV	Sufficiency NPV
AB no Gender	0.8665	1.028	0.9757	0.7810	1.0455

Table 12: Fairness metrics

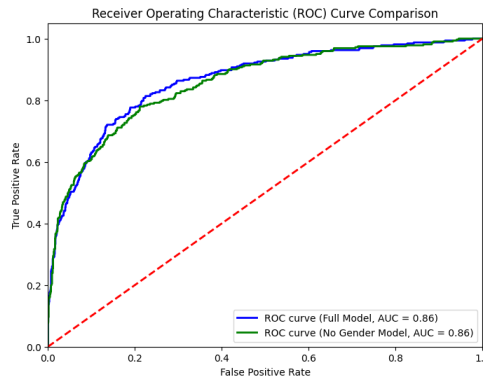


Figure 11: ROC Curve for Full and No Gender models.